

## Abstract

Ultrafast coherent phonon spectroscopy was performed in Tellurium. By using femtosecond light pulses at 10 kHz repetition rate, with 800 nm as a pump, and 1300 nm probe, the transient reflectance in the linear and SHG response was measured over a fluence scan of 0.48 to 4.69 mJcm<sup>-2</sup>. Rotational Anisotropy (RA) spectroscopy was implemented to symmetry-resolve the polarisation-dependence of the crystal structure [1]. The  $A_1$  coherent phonon mode was resolved in both linear and SHG responses, fitting its dynamics with the *Displacive Excitation of Coherent Phonons* (DECP) mechanism. In the SHG response, a potential pump-induced phase transition was explored [2].

## 1. Introduction

- Tellurium crystallizes in space group P<sub>3</sub>2<sub>1</sub> and point group 32 ( $D_3$ ), is birefringent and has a large SHG response ( $\chi^2 \approx 700 \text{ pmV}^{-1}$ ) in the near/mid-IR.
- Ultrafast pump-probe experiments in the linear response have demonstrated coherent phonon oscillations [3].
- Rotational Anisotropy (RA) spectroscopy has been proved to track light-induced phase transitions by crystal symmetries [1].

### DECP model

The transient response gets modulated by the coherent phonon dynamics:

$$\frac{\Delta I(\Delta t)}{I_0} = S(\Delta t, \sigma) \left[ B(\Delta t) - \sum_j A_j e^{-\Delta t/\tau_j} \cos(\omega_j \Delta t + \delta_j) \right] \quad (1)$$

With  $S(\Delta t, \sigma)$  the on-set pump and  $B(\Delta t)$  the background.

### Raman modulation

- The Raman-active phonon modes modulate the optical properties through the polarisation dependence.

$$\Delta \chi_{ijk}^{(2)}(t) = M_{il}(t) \chi_{ijk}^{(2)}(0) \quad (2)$$

$$M_{il}(t) = Q_A(t) R(A_{il}) + Q_E(t) [R(E_{xil}) \cos^2 \theta_p + R(E_{yil}) \sin^2 \theta_p]$$

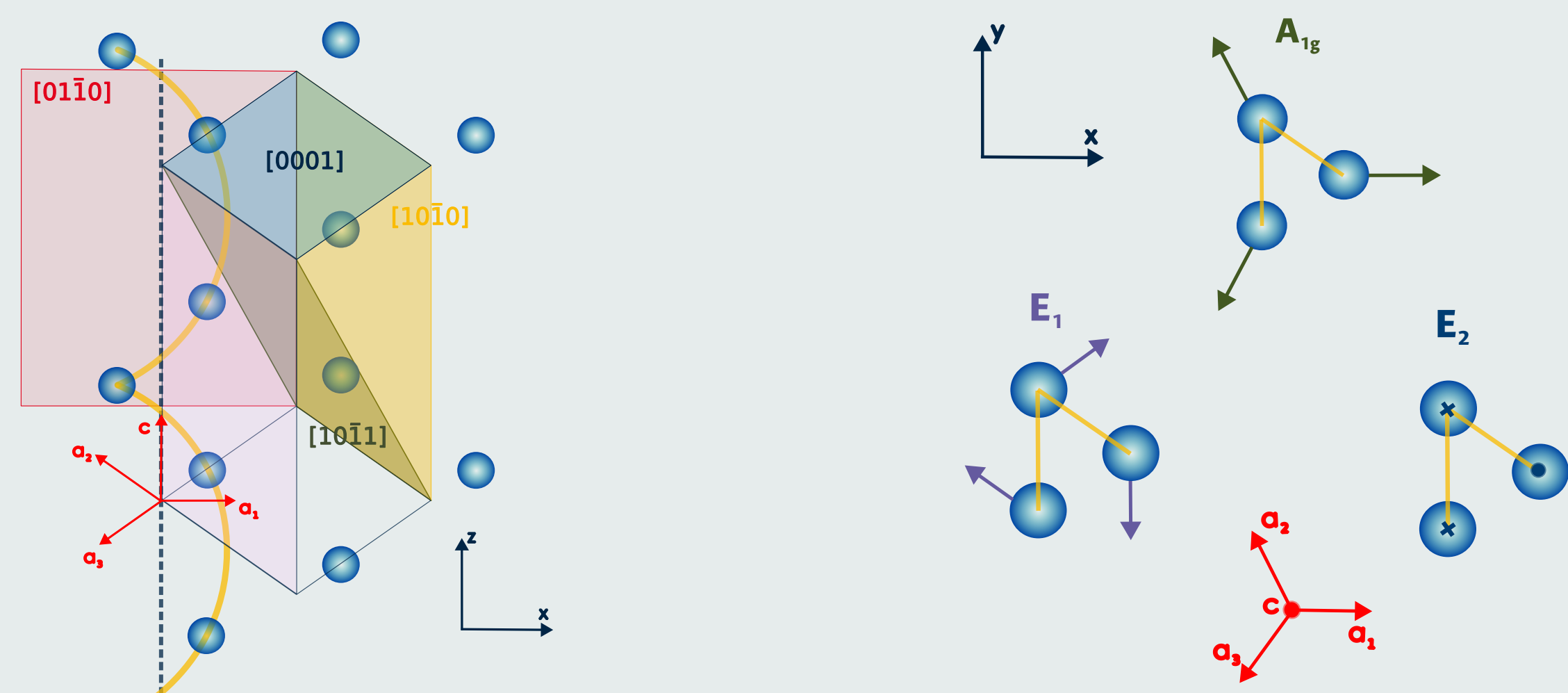


Figure 2. (Left) Crystal structure of Te. (Right) Coherent phonon modes.

## 2. RA and Pump-Probe Spectroscopy

- Ti:Sa laser at 10 kHz repetition rate,  $\sim 40$  fs pulses, with 800 nm pump and 1300 nm probe.
- Probe beam double-passes a rotating (5kHz) Half-Wave Plate (HWP), sampling 5 shots per angle.
- Tellurium sample hold in a cryostat at 90K and UHV conditions.
- Fundamental (NIR) and SHG reflection split to separate Photo-Diode (PD) detectors.
- Normalised transient reflectance:

$$\frac{\Delta I(\Delta t)}{I_0} = \frac{I_{\text{on}}(\Delta t) - I_{\text{off}}(\Delta t)}{I_{\text{off}}(\Delta t)} \quad (3)$$

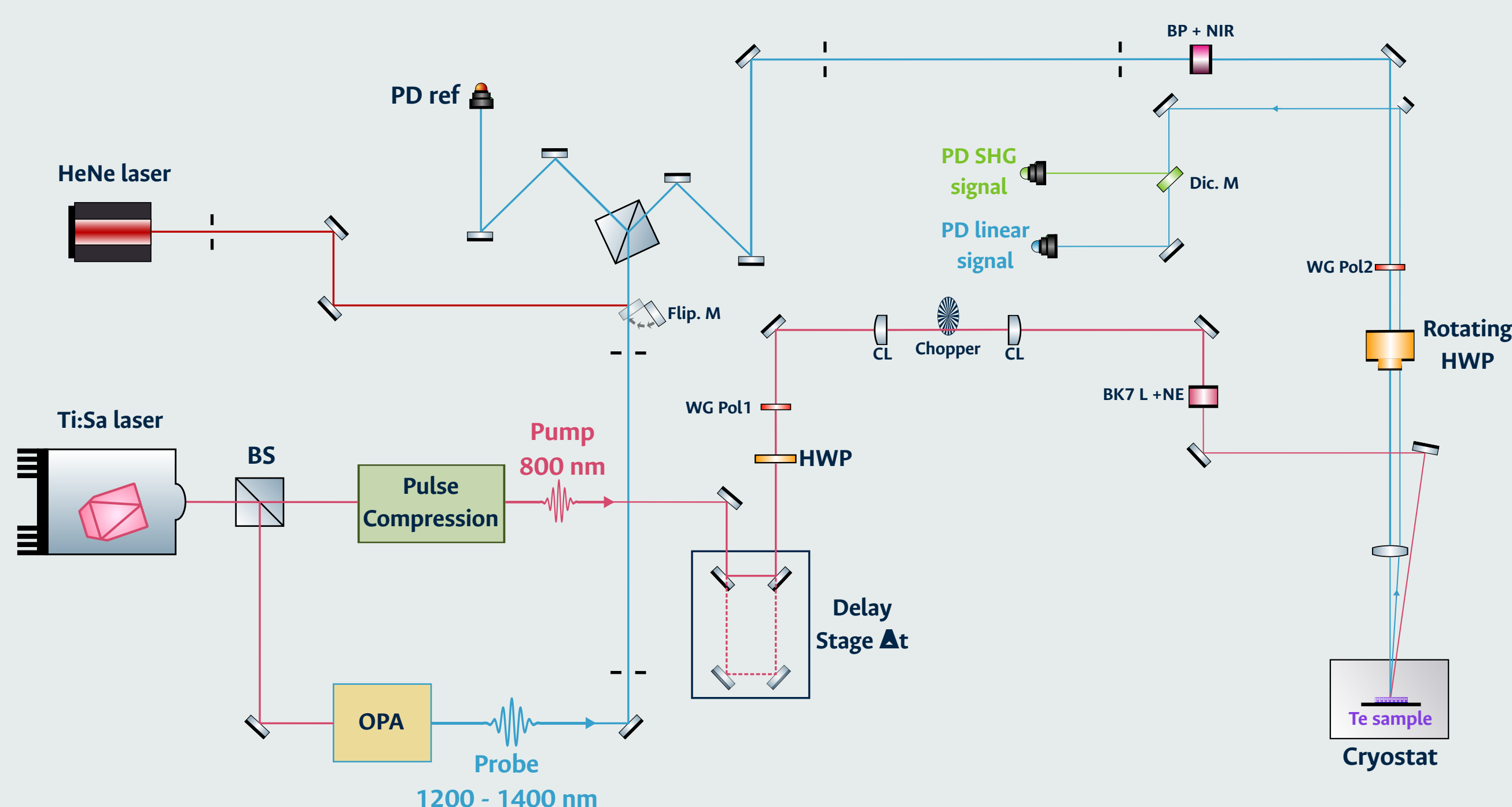


Figure 3. Experimental Setup.

## 3. Results

- Stripe pattern in the transient reflectance as the signature of coherent phonons.

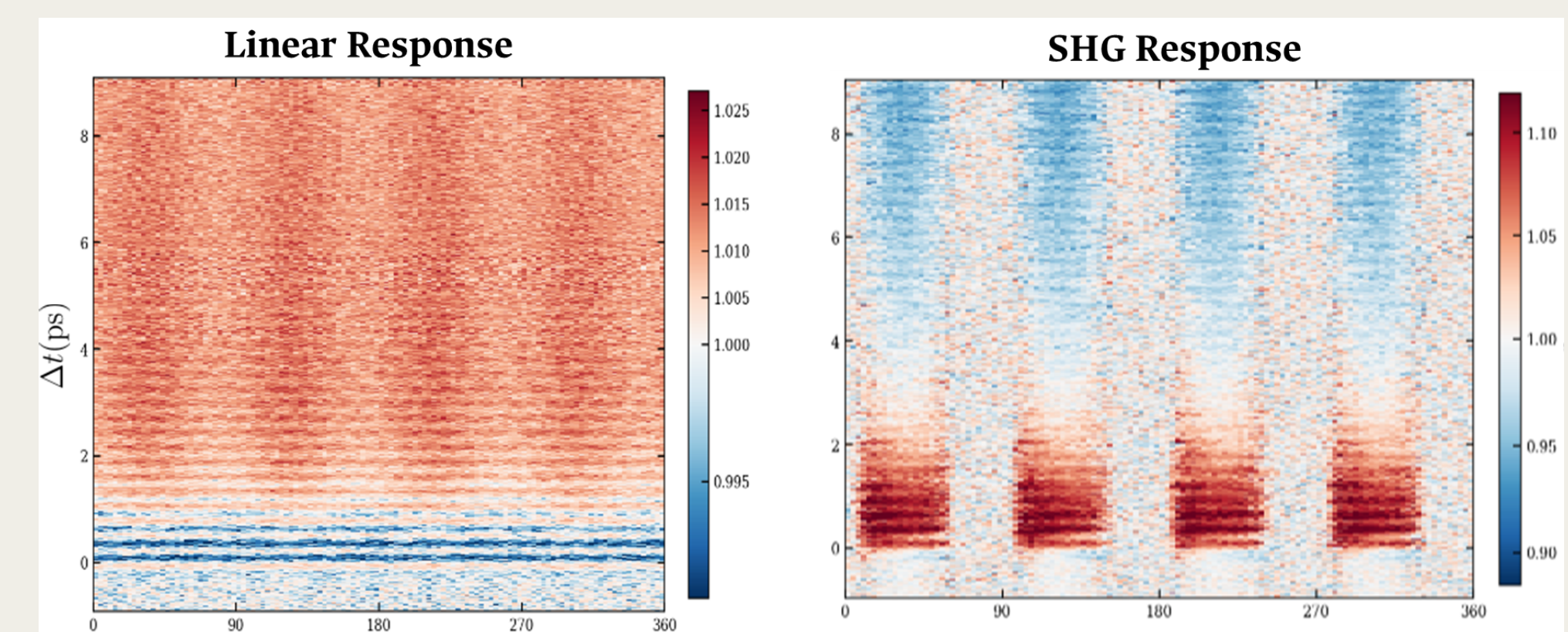


Figure 4. Pump-probe traces: polarisation- and time-resolved.

### Rotational Anisotropy Analysis

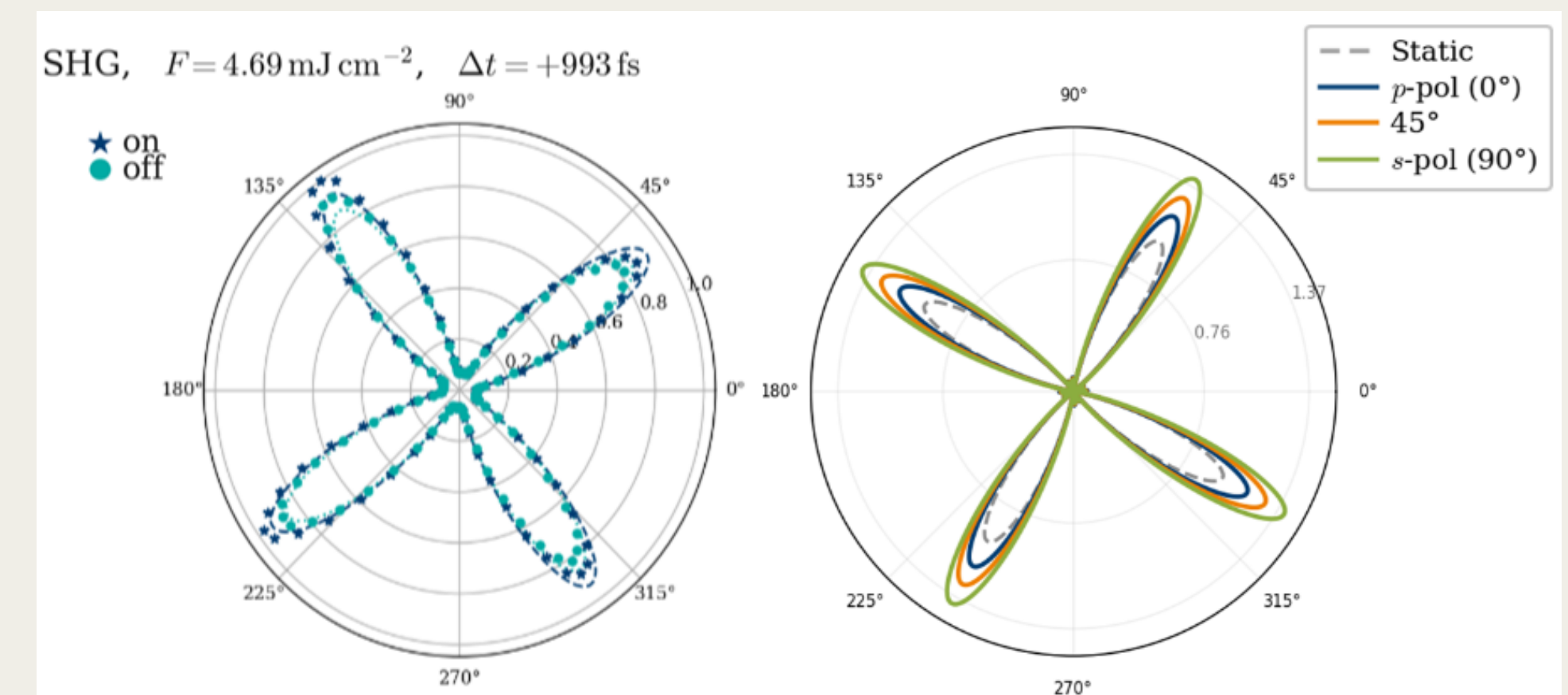


Figure 5. Rotational Anisotropy measured (left), and simulated (right) patterns.

- SHG polarisation dependence enabled a symmetry-resolved analysis of the crystal structure.

### Time-resolved and Spectral Analysis

- The transient reflectance is modulated by coherent phonons, fitted with the DECP model.

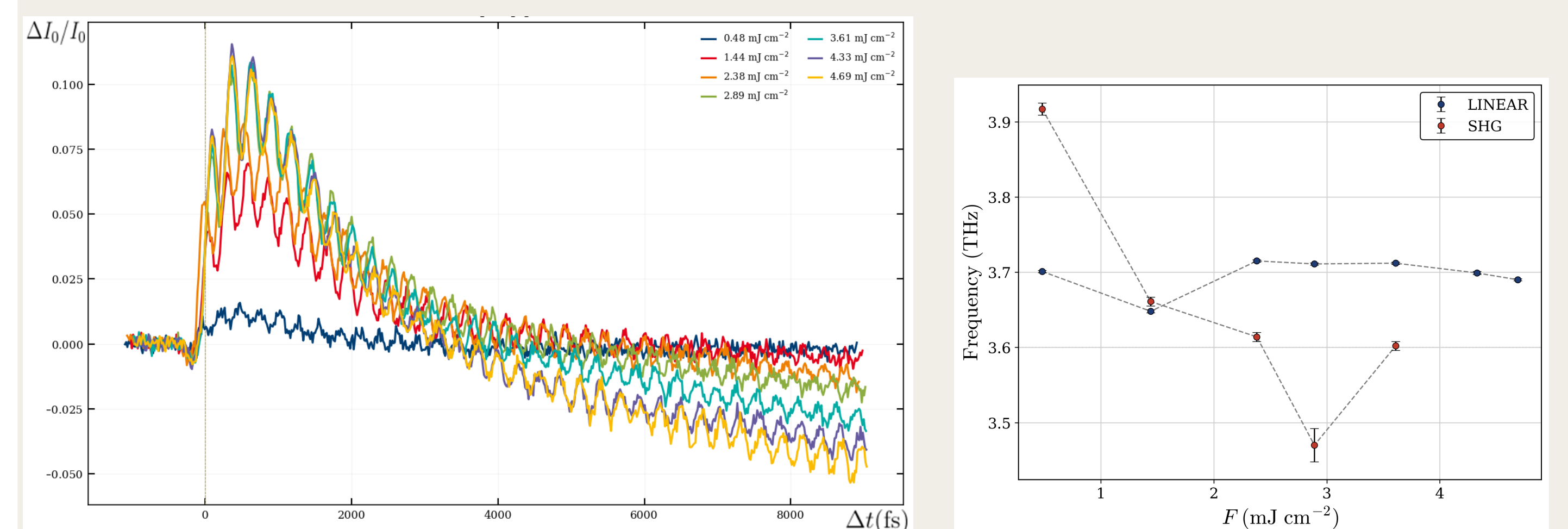


Figure 6. (Left) Coherent phonon oscillations. (Right) Fitted frequencies.

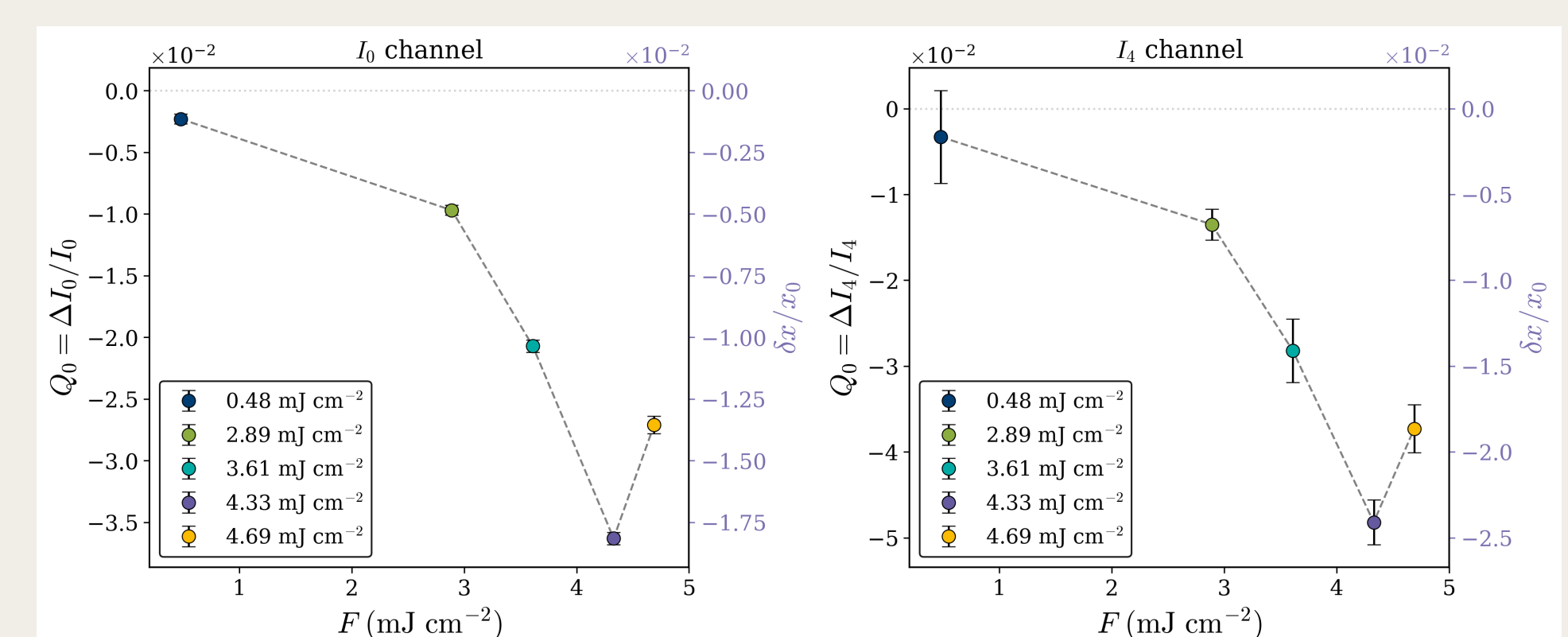


Figure 7. Late-time fitted amplitudes in the SHG response.

## 4. Outcomes and perspectives

- RA analysis as a powerful tool for symmetry-resolved ultrafast dynamics.
- $A_1$  phonon dynamics well fitted in the linear and SHG response.
- SHG response showed  $A_1$  frequency softening and an abrupt change in the late-time amplitude, a potential signature of a pump-induced phase transition.

## References

- [1] K. M. Siddiqui et al. 2025. DOI: 10.1063/5.0279973.
- [2] H. Ning et al. 2022. DOI: 10.1103/PhysRevB.106.205118.
- [3] A.-T. Kim et al. 2003. DOI: 10.1103/PhysRevB.68.012301.